

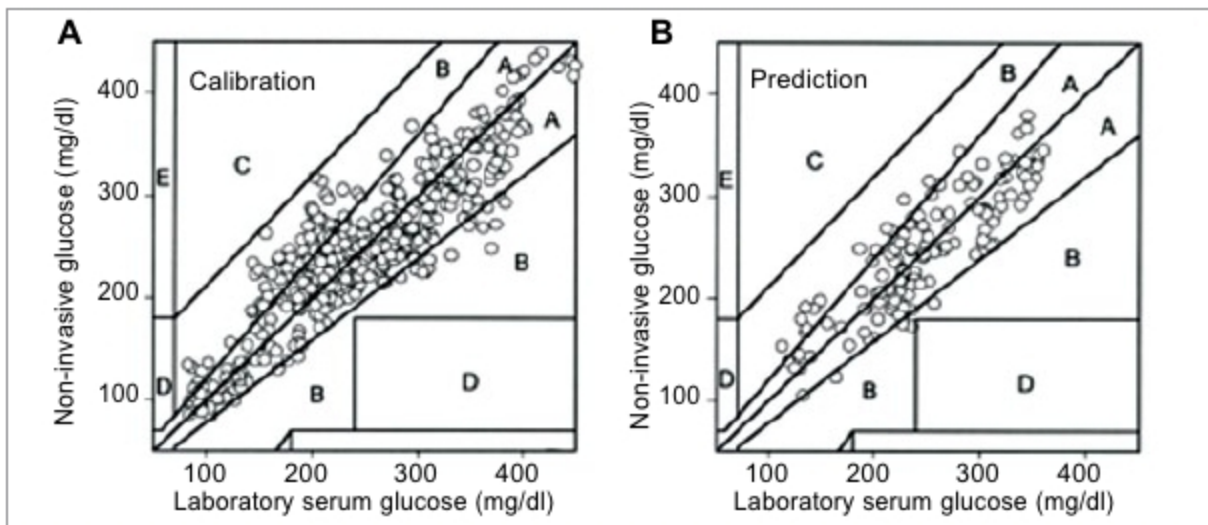


Fig. 9: Plot of laboratory versus true glucose data, SD(mg/dl), % mean absolute relative error (MARE) and correlation coefficient of r . Both plots show quality control invasive serum glucose calibration data, that is, invasive (laboratory) and non-invasive glucose concentration for calibration (A) and prediction (B)

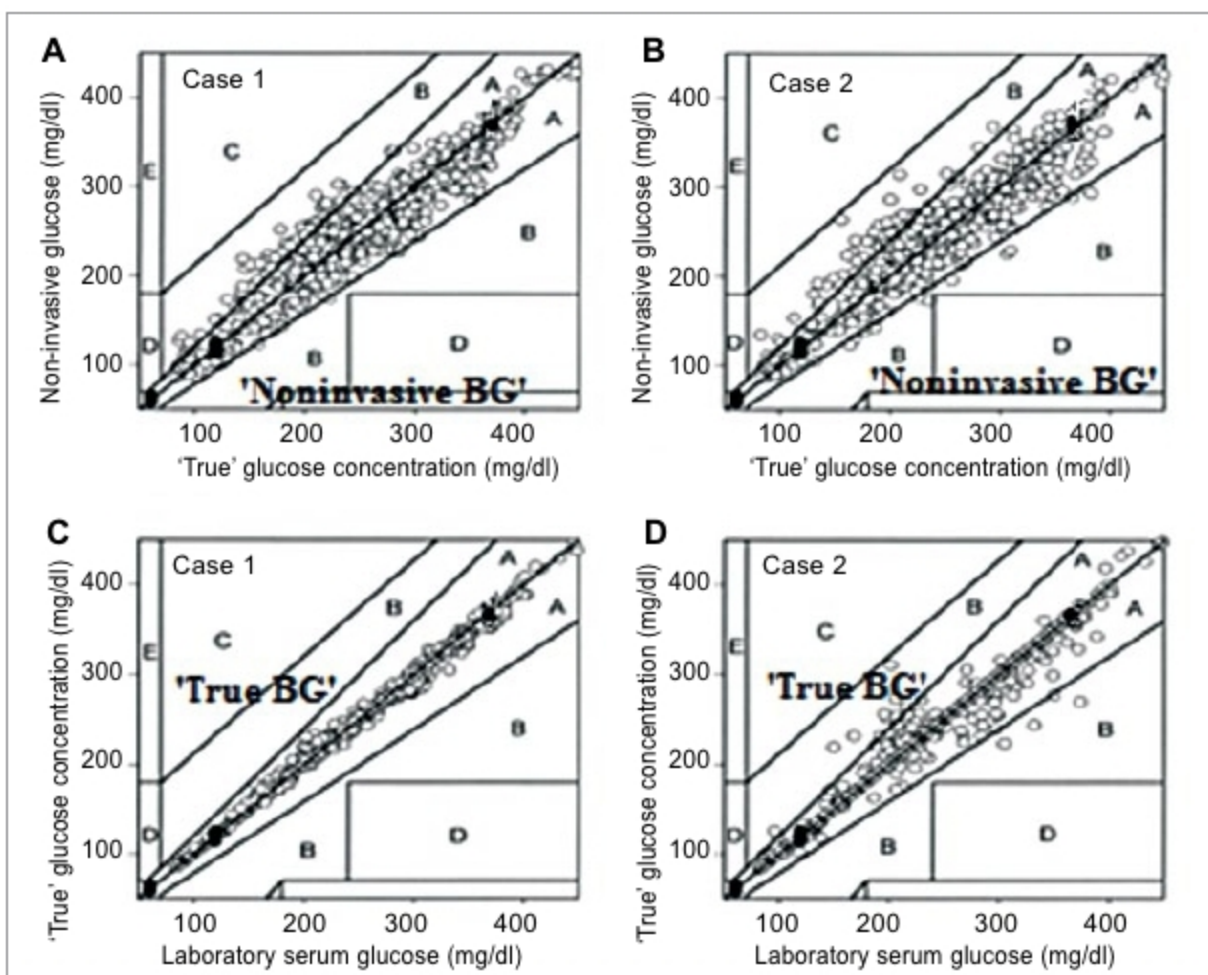


Fig. 10: Plot of correlations between true serum glucose invasive concentrations and non-invasive glucose concentrations for cases A and B, and corresponding plots of correlations between invasive (laboratory) serum glucose concentrations and true serum glucose concentrations for cases C and D

12 hours. The first clinical study with the new technology based on TES is promising, although the number of subjects tested is small.

Non-invasive measurement data analysis. In the following two cases of data analysis, orthogonal regression is used to introduce errors associated with the laboratory serum glucose method.

1. Clarke EGA - NIR spectroscopy renders relative errors of laboratory versus true glucose data, which were set at ± 3 per cent. This was provided for laboratory data from quality control measurements over the entire

sampling range of serum glucose concentrations. In clinically-accurate zone A, there is 89.4 per cent non-invasive data, and in clinically-acceptable zone B, there is 10.6 per cent data.

2. The orthogonal regression method calculates the best least-squares fit between laboratory and non-invasive measurements.

Deming regression has been widely adopted in clinical medicine because, unlike least-squares linear regression, this method only allows for imprecision in response variable y , whereas it can be present in both x and y variables.

Passing and Bablok's methods of comparison are also useful when imprecision occurs in both variables. However, imprecision can have a non-constant variance over the sampling range.

In clinically-accurate zone A, there is 97.5 per cent invasive measurement, and 2.5 per cent in clinically-acceptable zone B. There was a qualitative difference in the correlation plot of data points when arbitrary distribution of error for laboratory versus true glucose data were applied (cases C and D in Fig. 10). In this case, most laboratory versus true glucose concentrations lie on the identity line—97.5 per cent points deviating from this line are less than 20 per cent (case A), and around 2.5 per cent points deviating are greater than 20 per cent (case B). These results were calculated statistically, indirectly from orthogonal regression. These do not necessarily indicate that laboratory glucose data possess such errors.

Calibration of optical non-invasive measurements. Calibration of instruments and validation of results obtained by non-invasive optical methods *in vivo* under different environmental conditions and used by different patient populations, that is, of different ages, sizes and ethnic origins, must be performed. The devices may need to be precisely calibrated for individual users.

Current instrumentation requires specificity due to substantial chemical and physical interferences. The devices use multi-variate regression analyses that convert the optical signal to a glucose concentration (mg/dl). Large amounts of data are used to build the glucose model. This must take into consideration concentration range, sampling environment and other factors involved in the analysis. Therefore an instrument must be designed to detect accurate glucose concentration. Correlation and clinical interpretation of this value, with respect to the patient's true glucose value, is the dependency factor for optimum therapy in diabetes management.

Non-invasive glucose meters' current state. Looking at recent developments in healthcare technology,